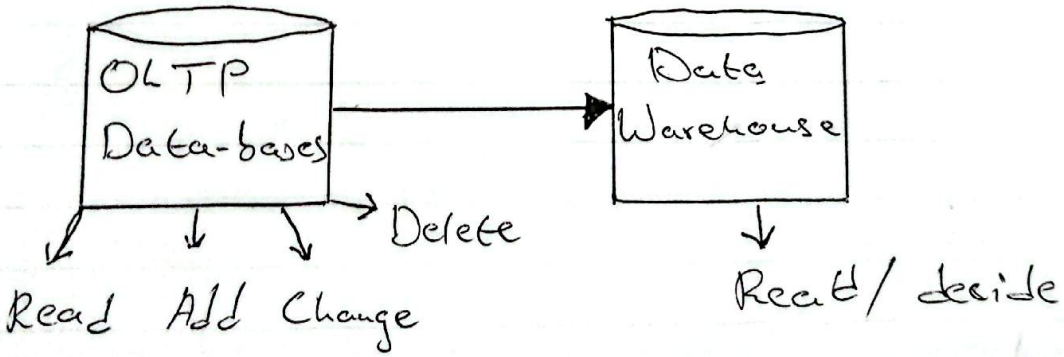
School Name: Bright learn
Student name: Sifiso Victor Masemola
Email: sifisovictormasemola@gmail.com
Date: 12 April 2026
Topic: Research assigment 1

① Database - is where data is stored.
①.② They are important for making sure that information does not get lost or mis-intepreted.
①.③ Databases are used in Corporate places and Industries e.g ((i) Vodacom (ii) Mr D

② Dataware-house - Is a centralized, digital repository that stores multiple or large data-base from multiple-sources.
②.① Dataware-house have larger ~~dat~~ storages than data-base.
②.② Companies use database warehouse for analytics and decision making.



Ref: Google www.geeksforgeeks.org

3) Data-lake → A centralize, Scalable repository that stores vast amounts of raw, unstructured and structured data in its native format.

3.1) Data-lakes → Stores vast amount of raw unstructured and structured data for data science and machine learning and Data ware-house → store processed, structured data for business intelligence and reporting.

(i) Data-lake → Sch Schema on-read

(ii) Data warehouse → Schema-on-write

3.2) Businesses would prefer data-lake to take unstructured data in-order to study, process it and store it according to the Business preferencess.

Ref: Google Chrome.

4) Data mart → Is a focused, subset of a data-warehouse designed for a specific business line, departments or teams.

4.1) Data mart is specific e.g department or usergroup while as data warehouse are centralized for the entire organizations data.

4.2) They are typically used by specific business departments or teams within an organization

(i) Marketing

(ii) HR

(iii) Finance

Ref: Google chrome

(5.) Structured data → data-stored in table with rows and columns e.g Spreadsheet.

(5.1) Semi-Structured data → a form of information that does not reside in a grid relational database but uses tags and enforce hierarchies

(5.2) Unstructured-data → It does not have consistent format e.g Music, Videos and phone calls.

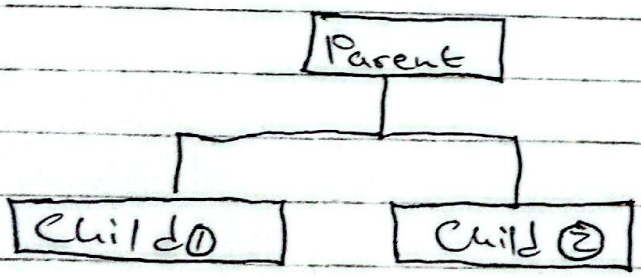
Ref: Google chrome & Class notes

(6) Data Modelling ⇒ Process of creating visual, structured blueprints (diagrams, symbols) that define how data is collected, used ~~use~~ within information system.

(6.1) Important because, it helps in organizing and structuring data in a way that is easy to understand and manage

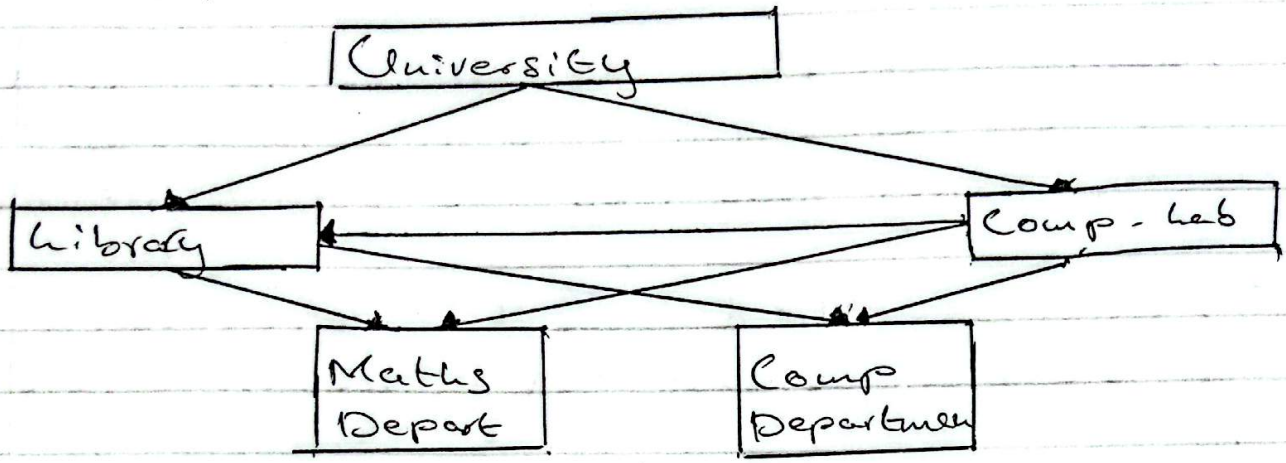
Ref: www.Urbanpro.com

(6.2) Hierarchical model → organize data in a tree form, parent-to child structure



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(6.2) Network modelling → represents data using a flexible graph structure



Ref: educba.com

(7) Data-mining → Process of analyzing large, complex datasets to discover hidden patterns, correlations and anomalies transforming raw data into actionable insights.

(7.1) Data - Mining → is used to analyse massive datasets using statistics and database system to identify hidden patterns.

(7.2) Financial fraud detection and retail market basket analysis.

Ref: google.com

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(8) Database management system → Is a system or a device you can use to access data. it can be used for efficient creation, retrieval, updating and delete data.

(i) Sales IQ → Live chat app to engage and convert website visitors.
(ii) Webinar → For webcasting online webinars.
(iii) Landing page → Smart landing page builder to increase conversation rates.
(iv) Page sense → Website conversation optimization and personalization platform.

Ref: www.zoho.com

(9) ETL → data intergration process that gathers raw data from multiple sources, converts it into a structured format, and loads it into a centralized system like a data-warehouse.

Ref: Amazon web service

(9-1) Extract: Collection of data.
9.2 Transform: Reading and processing of raw data.
9.3 Load: Visualize and assemble data.
9.4 ETL is important because it enables organizations to intergrate data sources into a unified repository for actionable insights.

Ref: Google.com

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(10) XML → Extensible Markup language
Structured data format favored for complex hierarchical documents

10.2 TXT → Plain text Simplest format, storing unformatted text

10.3 ORC → Optimized Row columnar faster for writing and suited for Apache Hive/Hadoop

Ref: Google.com

(11) Artificial Intelligence (AI) → Is a set of technologies that empowers computers to learn, reason and perform a variety of advanced tasks in ways that used to require human intelligence.

11.2 Narrow AI: is designed to perform specific, predefined tasks within a limited context.

11.3 General AI: Is a theorical, human-like intelligence capable of applying knowledge across any domain. It does not yet exist.

(12) Machine learning (ML) → Is a AI that develops algorithms to identify pattern in data, allowing computers to learn and make predictions without being explicity programmed.

Ref: google.com

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(12.) Traditional programming human write the rules while in machine learning, the computer learns them from data.

(i) Supervised learning	Unsupervised
* Labeled	* Unlabeled
* Predict outcomes for new data	* Discover hidden patterns or structures
* Generally simpler	* More complex
* High validated against ground truth	* Variable, requires human interpretation to validate.

(13) Deep learning → a subset of machine learning inspired by the structure of the human brain.

(13.1) Deep learning is a specialized subset of machine learning (ML), which is a branch of AI. While AI represents the broad goal of creating human-like intelligence

(13.2) Neural networks → Computational models inspired by the human brain that use layers of interconnected nodes.

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(14) Natural language Processing (NLP)

* It is a branch of artificial intelligence that allows computers to understand, interpret and generate human language

(i) Health care

(ii) Finance

iii) Customer experience

Ref: Google.com

(15) large language Model (LLM)

* It is a specialized type of AI trained to deal with large amount of data to understand, generate and manipulate human language.

Ref: Google.com

Feature	TML	LLM
Data type	* Tables, numbers, precise labels	* Texts, articles, websites, code
Scope	* Task specific: built for one job eg fraud detection, image labeling	* Versatile model can summarize, translate, code and chat
Output	Predictions or labels eg True/False or price forecast	Human-like text full sentences, paragraphs or creative content
Resources	* Can often run on standard computers or local servers	* Require massive computing power (thousands of GPUs/TPUs)

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(16) Generative AI → a branch of artificial intelligence that can create entirely new content, such as text, images, audio, video and computer code.

(16.1) Generative AI → It uses large datasets to learn the underlying patterns, structures and statistical relationships of information.

(16.2.1) Chatbots → They are designed to understand and generate human-like responses to text-based, Natural language prompts.

(16.2.2) AI meeting assitant (AI note takers)

* Can transcribe and summerize online meetings and have the potential to stifle conversation and open inquiry.

(16.2.3) API → Application Programming Interface

Enables developers to intergrate large language Models (LLM) into their own applications, products or services.

Ref: huit.harvard.edu

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(17) Training dataset → a massive collection of text, images or code used to teach an AI model how to recognise patterns or generate a new content.

(17.1) It is the foundation upon which any AI or machine learning model is built.

(17.2) AI model will inevitably fail - a concept famously known as (GiGO). Because the model learns entirely from the patterns in that data any flaws in the "textbook" become permanent flaws in the "student."

(i) Model hallucinations
(ii) Algorithmic bias and discrimination
(iii) Overfitting and underfitting
iv) Security Vulnerabilities (Data poisoning)
v) Concept drift.

Ref: Google.com

(18) Computer Vision → a field of artificial intelligence that enables computers and systems to derive meaningful information from digital images, videos and other visual inputs.

(18.1) Machine learning works by finding mathematical patterns in data, turning data into numbers eg 0:1

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(i) Biometrics
(ii) Sports e.g FIFA VAR

Ref: google.com

(19) API → Application Programming Interface
* Is a set of rules and protocols that allows different software applications to communicate, exchange data and share functionality.

(19.1) In a restaurant (Steers, KFC, Debonairs etc) you use a menu to choose from available items and give your order and the kitchen (the server) brings back your food.

(19.2.1) They play Intermediary role in allowing systems to interact without needing to know how the other is built.

(19.2.2) Security and efficiency - Only share necessary data, protecting internal systems and enhancing security.

Ref: Google.com

(20) API key → a unique alphanumeric string (like a password) that identifies and authenticates an application or project making a request to an API.

(20.1) The core purposes are, Identification, Access control, Usage tracking and rate limits.

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(20.2) API keys should be stored outside of your source code to prevent accidental exposure especially in version-controlled systems like Github.

(21) Rest API → An architectural style for designing web services that allows different systems to communicate over the internet.

(21.1) Rest stand for Representational ~~Stand~~ State Transfer

(21.2)

Method	Action	Exemple
GET	Retrieve data	Get list of users
POST	Create new data	Add a new user
Put	Replace/Update data	update user information
Delete	Remove data	Delete user information

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Brightlearn Assignment

(22) What is prompt Engineering.

* The art and science of designing, structuring and refining inputs to guide GAI models toward producing accurate, relevant and high-quality outputs.

(22.1) It is commonly used with API to bridge the gap between open ended user intent and the specific, structured requirements of high-scale software applications.

(22.2) json (Java Script Object Notation)

```
{
"id": 101,
"name": "Alex Rivera",
"is_active": true,
"roles": ["admin", "editor"]
}
```

Ref: Google.com

(23) It is important to have a skill, with chatGPT or Claude because it is becoming a fundamental professional literacy much like computer literacy was in previous decades.

Poor prompt	Upgraded prompt
* "Write a python script to check a website."	* "Write a python script using the requests library to check if https://google.com returns

Ref: Google.com	Print the error code. Include comments explaining each line and format the output as a clean function"

(24) MAP function → a tool that takes a list of items and applies the same change to every single one of them creating a brand-new list with the results.

(24.1) It takes an input collection and a function as arguments. It applies that function to each element and returns a new collection of the same size.

(25) Cloud computing → is the on demand delivery of IT resources like servers, storage, databases and software - over the internet with pay-as-you-go pricing.

(25.1) Saas (Software as a service) → ready to use applications delivered over a browser e.g Gmail, Salesforce and Slack.

25.2 Paas (Platform as a service) → Provides a framework for developers to build, test and deploy apps without worrying about the underlying hardware e.g Google, App engine and Heroku.

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(25.3) Iaas (Infrastructure as a Service) → The most flexible model, giving you "raw" computing building blocks like virtual servers and storage e.g AWS EC2, DigitalOcean.

(26) Version control system → is a tool that tracks and manages changes to software code and files over time

(26.1) Git - a distributed version control system designed to track changes in source code and files during software development or data analysis projects.

(26.2) Developers and analysts uses it to take snapshots of their work, revert to previous versions if mistakes are made and collaborate simultaneously without overwritting each other's work.

(26.3.1) Commit → is a permanent snapshot of your project files at a specific point in time.

(26.3.2) Branch → is an independent line of development that allows you to work on new features or fixes without effecting the "main" code.

Ref: Google.com

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(27) Jupyter Notebook → an open-source, browser-based application used to create and share documents containing live code, visualizations and narrative text.

(27.1) Python is the most common and dominant language used with jupyter Notebook.

(27.2) Is stable for data scientists and analysts because, it bridges the gap between raw code and human-readable narrative, facilitating an iterative "Conversation" with data.

(28) What is Python → is an interpreted object-oriented, high level programming language with dynamic semantics. Its high level built in data structures. Ref: Python.org

(28.1) It is popular to AI and data analytics because it is a powerful ecosystem, good accesibility and strong community.

(28.2.1) Pandas → Primary tool for data manipulation analysis and cleaning

(28.2.2) Numpy → The foundation for numerical computing and handling large arrays.

(28.2.3) Scipy → Essential for scientific computing and advanced algorithms

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Bright learn Assignment

(28.2.4) Polars → A fast dateframe library same as Pandas.

(29) Data Privacy → The right of individuals to control how their personal data - such as identifiers, behavior or sensitive information is collected, used, shared and stored by organizations. Ref: Google.com

(29.1) It is important because an Individual has a right to decide how data is collected, store and used. To avoid criminal activities and victimizing a person.

(29.2) GDPR (General Data Protection Regulation)
→ A comprehensive EU law, effective since May 25, 2018, that protects the personal data and privacy of individuals within the EU and EEA.

(29.3) Countries that use GDPR are Germany, France, Italy, Spain, and Ireland.

(30) Data Ethics → is a branch of ethics that evaluates the moral implications of data collection, processing, sharing and usage particularly regarding AI and algorithms.

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30.1) Exploitation → AI models can be
* developed by exploiting, underpaying or exploiting the data of ~~them~~ vulnerable individuals or marginilized populations

30.1.2) Academic and creative dishonesty
* Using AI to research or art without proper attribution undermines human creativity and intellectual intergrity.

30.1.3) Privacy and Surveillance → The use of AI-powered facial recognition by governments and companies in public spaces enables mass surveillance, tracking individuals without consent.

Ref: Google.com

31) Data governance → a structured framework of policies, processes, roles and technology used to manage an organizations data assets securely, accurately and efficiently throughout their lifecycle.

31.1) Policies and Standards ÷ Setting internal rules for how data is collected, stored, processed and disposed of.

31.2) Roles and responsibilities
* Defining accountability, such as data stewards and owners, to ensure proper handling.

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3.1.3) Security and Privacy - Restricting access and protecting sensitive information to meet regulatory requirements like GDPR or HIPAA.

3.1.4) Data Quality → Establishing standards for accuracy, completeness and consistency.

3.2) Bias in AI → refers to the occurrence of biased results due to human biases that skew the original training data or AI algorithm - leading to distorted outputs and potentially harmful outcomes.

Ref: Ibm.com

3.2.1) When its outputs systematically produce unfair, prejudiced or inaccurate results that favor certain groups or perspectives over others.

Ref: Sap.com

3.2.2.1) Source of bias: Primarily arises from skewed incomplete data.

3.2.2.2) Impact: Leads to discrimination and reinforces stereotypes such as mis-identifying faces or associating specific genders with particular professions.

3.2.2.3) Measurement bias - Errors in how data is collected or measured, such as defaulting to gendered roles

32.2.4 Sampling bias - The training data does not represent the real world scenario accurately.

32.2.5 Stereo typing - The model makes Prejudiced assumptions, such as defaulting to gendered roles

32.2.6 Algorithmic - The model's internal design or optimization choices amplify discrepancies.

33 key differences between, data analyst, Data scientist, data engineer and AI engineer

Features	Data analyst	Data scientist	Data Engineer	AI Engineer
Roles	Interpreting past data & reporting	Predicting future trends & advanced modelling	Building data pipelines	Deploying operationalizing AI models
Skills	SQL, Excel, visualization	Python / R, Stats, Machine learning	SQL, Spark, ETL/ELT	Python, LLM's, API's
Tools	Tableau, Power BI, Excel SQL	Scikit-learn, Pandas, Jupyter Notebooks	Snowflake, Spark, Kafka	Open AI, API, Weights, Biases

Ref: Youtube / Google Com

Bright learn

(54) Business Intelligence → is an umbrella term for the strategies and technologies used by enterprises for the data analysis of business information Ref: Microsoft.com

34.1) Qlik sense → good for complex data exploration

34.2) Zoho analytics → affordable and quick intergration with the zoho ecosystem.
Ref: Oval edge.com

(55) Firstly I would like to thank the platform for a wonderful experience. Section A was suprising that I missed out a lot since I am an IT graduate and it feels like I was not taught enough. It makes sense now.

To be honest I found themall interesting but the one I found most interesting is Section C because my previous school didnot teach me that much of coding and I was taught OOP (object Oriented programming) and I loved it but I had little exposure to it.

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(36) The difficult concept for me was Section B and D because AI is a subject I never touched on in school, so it is very new to me. Google and Youtube gave me a light on the subject but I still feel I need more time to learn.

(37) In future I see myself ending up as a back-end developer after some studies I made. Youtube video by Coursera

In future engineers will be more essential than data analysts. AI will replace Might data analysts especially if it is provided with complete and accurate data.

With that said I plan to start as a data analyst to get to understand the field more and the more I get used to it, focus more in converting to a engineer.

38. References

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